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LAPORAN JOBSHEET 14

Implementasi ArrayList

➤ Kode Program Customer15

```
public class Customer15 {  
    public int id;  
    public String name;  
  
    public Customer15()  
    {  
  
    public Customer15(int id, String name) {  
        this.id = id;  
        this.name = name;  
    }  
  
    public String toString() {  
        return " ID: " + this.id + " Nama: " + this.name;  
    }  
}
```

➤ Kode Program Book15

```
public class Book15 {  
    public String isbn;  
    public String title;  
  
    public Book15 (String isbn, String title) {  
        this.isbn = isbn;  
        this.title = title;  
    }  
  
    public String toString() {  
        return "ISBN: " + this.isbn + " Title: " + this.title;  
    }  
}
```

➤ Kode Program DemoArrayList15

```
import java.util.ArrayList;
```

```

public class DemoArrayList15 {
    public static void main(String[] args) {
        ArrayList <Customer15> customers = new ArrayList <> (2);

        Customer15 customer1 = new Customer15(1, "Zakia");
        Customer15 customer2 = new Customer15(5, "Budi");

        customers.add(customer1);
        customers.add(customer2);
        customers.add(new Customer15(4, "Cica"));

        for (Customer15 cust : customers) {
            System.out.println(cust.toString());
        }
    }
}

```

➤ **Hasil Program**

ID: 1 Nama: Zakia ID: 5 Nama: Budi ID: 4 Nama: Cica

Compile dan run kode program, di mana object yang baru ditambahkan? Di awal, di tengah, atau di akhir collection?

Jawab: Berada di akhir collection.

➤ **Kode Program tambahan DemoArrayList15**

```
customers.add(2, new Customer15(100, "Rosa"));
```

➤ **Hasil Program**

ID: 1 Nama: Zakia ID: 5 Nama: Budi ID: 100 Nama: Rosa ID: 4 Nama: Cica

Compile dan run kode program. Index pada ArrayList dimulai dari 0 atau 1?

Jawab: Index pada ArrayList dimulai dari 0.

➤ **Kode Program tambahan DemoArrayList15**

```
System.out.println(customers.indexOf(customer2));
```

➤ **Hasil Program**

```
1
ID: 1 Nama: Zakia
ID: 5 Nama: Budi
ID: 100 Nama: Rosa
ID: 4 Nama: Cica
```

➤ **Kode Program tambahan DemoArrayList15**

```
Customer15 customer = customers.get(1);
System.out.println(customer.name);
customer.name = "Budi Utomo";
```

➤ **Hasil Program**

```
1
Budi
ID: 1 Nama: Zakia
ID: 5 Nama: Budi Utomo
ID: 100 Nama: Rosa
ID: 4 Nama: Cica
```

Cobalah hapus angka 2 saat instansiasi object customers. Apakah ArrayList dapat diinstansiasi tanpa harus menentukan size di awal?

Jawab: Ya, ArrayList dapat diinstansiasi tanpa harus menentukan size di awal.

➤ **Kode Program tambahan DemoArrayList15**

```
ArrayList <Customer15> newCustomers = new ArrayList <> ();
```

```
newCustomers.add (new Customer15(201, "Della"));
newCustomers.add (new Customer15(202, "Victor"));
newCustomers.add (new Customer15(203, "Sarah"));
```

```
customers.addAll (newCustomers);
```

➤ **Hasil Program**

```
1
Budi
ID: 1 Nama: Zakia
ID: 5 Nama: Budi Utomo
```

ID: 100 Nama: Rosa ID: 4 Nama: Cica ID: 201 Nama: Della ID: 202 Nama: Victor ID: 203 Nama: Sarah
--

Implementasi Stack

➤ Kode Program StuckDemo15

```
public class StackDemo15 {  
    public static void main(String[] args) {  
        Book15 book1 = new Book15("1234", "Dasar Pemrograman");  
        Book15 book2 = new Book15("7145", "Hafalah Shalat Delisa");  
        Book15 book3 = new Book15("3562", "Muhammad Al-Fatih");  
  
        Stack <Book15> books = new Stack<>();  
        books.push(book1);  
        books.push(book2);  
        books.push(book3);  
  
        Book15 temp = books.peek();  
        if (temp != null) {  
            System.out.println(temp.toString());  
        }  
        Book15 temp2 = books.pop();  
        if (temp2 != null) {  
            System.out.println(temp2.toString());  
        }  
    }  
}
```

➤ Hasil Program

ISBN: 3562 Title: Muhammad Al-Fatih ISBN: 3562 Title: Muhammad Al-Fatih
--

Mengapa perlu ada pengecekan (temp != null)?

Jawab: pengecekan (temp != null) diperlukan untuk mencegah program error ketika objek kosong, sehingga toString() hanya dijalankan jika variable temp memiliki data.

➤ **Kode Program tambahan StuckDemo15**

```
for (Book15 book : books) {  
    System.out.println(book.toString());  
}
```

➤ **Hasil Program**

ISBN: 3562 Title: Muhammad Al-Fatih ISBN: 3562 Title: Muhammad Al-Fatih ISBN: 1234 Title: Dasar Pemrograman ISBN: 7145 Title: Hafalah Shalat Delisa
--

Bagaimana cara melakukan pencarian elemen pada stack menggunakan method search()?

Jawab:

➤ **Kode Program tambahan StuckDemo15**

```
int posisi = books.search(book1);  
System.out.println("Buku ini berada di posisi " + posisi);
```

➤ **Hasil Program**

ISBN: 3562 Title: Muhammad Al-Fatih ISBN: 3562 Title: Muhammad Al-Fatih ISBN: 1234 Title: Dasar Pemrograman ISBN: 7145 Title: Hafalah Shalat Delisa Buku ini berada di posisi 2

Implementasi TreeSet

➤ **Kode Program TreeSetDemo15**

```
import java.util.TreeSet;  
  
public class TreeSetDemo15 {  
    public static void main(String[] args) {  
        TreeSet <String> fruits = new TreeSet<>();  
  
        fruits.add("Mangga");  
        fruits.add("Apel");  
        fruits.add("Jeruk");
```

```

        fruits.add("Jambu");

        for (String temp : fruits) {
            System.out.println(temp);
        }
    }
}

```

➤ **Hasil Program**

```

Apel
Jambu
Jeruk
Mangga

```

Compile dan run program. Mengapa urutan yang ditampilkan berbeda dengan urutan penambahan data ke dalam TreeSet fruits?

Jawab: Urutannya berbeda karena TreeSet mengurutkan data string berdasarkan alphabet.

➤ **Kode Program tambahan TreeSetDemo15**

```

System.out.println("First: " + fruits.first());
System.out.println("Last: " + fruits.last());
fruits.remove ("Jeruk");
System.out.println("Setelah remove " + fruits);
fruits.pollFirst ();
System.out.println("Setelah poll first " + fruits);
fruits.pollLast ();
System.out.println("Setelah poll last " + fruits);

```

➤ **Hasil Program**

```

Apel
Jambu
Jeruk
Mangga
First: Apel
Last: Mangga
Setelah remove [Apel, Jambu, Mangga]
Setelah poll first [Jambu, Mangga]
Setelah poll last [Jambu]

```

Apa yang dilakukan oleh method first(), last(), remove(), pollFirst(), dan pollLast()?

Jawab:

- First() : Mengambil elemen pertama
- Last() : Mengambil elemen terakhir
- Remove() : Menghapus elemen tertentu
- pollFirst() : Menghapus elemen pertama
- pollLast() : Menghapus elemen terakhir

Sorting

➤ Kode Program DemoArrayList15

```
import java.util.ArrayList;
import java.util.Collections;
```

```
public class DemoArrayList15 {
    public static void main(String[] args) {
        ArrayList <Customer15> customers = new ArrayList <> ();
        Customer15 customer1 = new Customer15(1, "Zakia");
        Customer15 customer2 = new Customer15(5, "Budi");

        customers.add(customer1);
        customers.add(customer2);
        customers.add(new Customer15(4, "Cica"));
        customers.add(2, new Customer15(100, "Rosa"));

        System.out.println(customers.indexOf(customer2));

        Customer15 customer = customers.get(1);
        System.out.println(customer.name);
        customer.name = "Budi Utomo";

        ArrayList <Customer15> newCustomers = new ArrayList <> ();
        newCustomers.add (new Customer15(201, "Della"));
        newCustomers.add (new Customer15(202, "Victor"));
        newCustomers.add (new Customer15(203, "Sarah"));
```

```
customers.addAll (newCustomers);
```

```
for (Customer15 cust : customers) {  
    System.out.println(cust.toString());  
}
```

```
ArrayList <String> daftarSiswa = new ArrayList<>();  
daftarSiswa.add("Zainab");  
daftarSiswa.add("Andi");  
daftarSiswa.add("Rara");  
Collections.sort(daftarSiswa);
```

```
System.out.println(daftarSiswa);
```

```
customers.sort((c1, c2) -> c1.name.compareTo(c2.name));  
}  
}
```

➤ Hasil Program

```
1  
Budi  
ID: 1 Nama: Zakia  
ID: 5 Nama: Budi Utomo  
ID: 100 Nama: Rosa  
ID: 4 Nama: Cica  
ID: 201 Nama: Della  
ID: 202 Nama: Victor  
ID: 203 Nama: Sarah  
[Andi, Rara, Zainab]
```